

Low-Power 12-Bit Analog-to-Digital Converter with I²C Interface

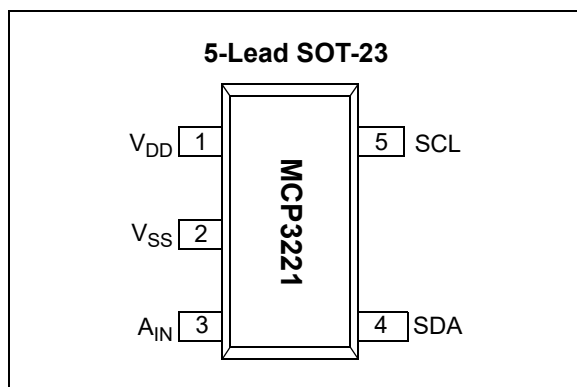
Features

- 12-bit Resolution
- ± 1 LSB Differential Nonlinearity (DNL) Maximum
 ± 2 LSB Integral Nonlinearity (INL) Maximum
- 250 μ A Max Conversion Current
- 5 nA Typical Standby Current, 1 μ A maximum
- I²C Compatible Serial Interface:
 - 100 kHz I²C Standard mode
 - 400 kHz I²C Fast mode
- Up to Eight Devices Connected on a Single 2-Wire Bus
- 22.3 ksp/s in I²C Fast mode
- Single-Ended Analog Input Channel
- On-Chip Sample and Hold
- On-Chip Conversion Clock
- Single-Supply Specified Operation: 2.7V to 5.5V
- AEC-Q100 Qualified, Grade 1
- Extended Temperature Range: -40°C to +125°C
- Package: 5-Lead SOT-23

Applications

- Automotive
- Data Logging
- Multi-Zone Monitoring
- Handheld Portable Applications
- Battery-Powered Test Equipment
- Remote or Isolated Data Acquisition

Package Types (top view)



General Description

MCP3221 is a successive approximation Analog-to-Digital Converter (ADC) with 12 bits of resolution. It provides one single-ended input with very low power consumption. Using advanced CMOS technology, MCP3221 provides a low maximum conversion current of 250 μ A and a standby current of 1 μ A.

The low current consumption and compact SOT-23 package make this device ideal for battery-powered and remote data acquisition applications.

MCP3221 communicates over a 2-wire I²C compatible interface. Standard (100 kHz) and Fast (400 kHz) I²C modes are available. An on-chip conversion clock enables independent timing for the I²C and conversion clocks. Up to eight MCP3221 devices can interconnect on a single 2-wire bus.

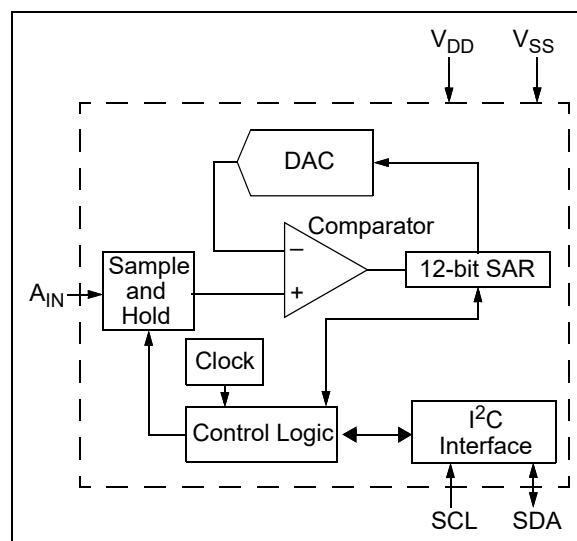
MCP3221 runs on a single-supply voltage operating over a broad range of 2.7V to 5.5V. The device offers excellent linearity: ± 1 LSB DNL and ± 2 INL, maximum.

MCP3221 is AEC-Q100 qualified for automotive applications and operates over the extended temperature range of -40°C to +125°C.

Related Devices

- MCP3021 – Low-Power 10-Bit Analog-to-Digital Converter with I²C Interface

Functional Block Diagram



MCP3221

1.0 ELECTRICAL SPECIFICATIONS

1.1 Absolute Maximum Ratings^(†)

V_{DD}	+7.0V
Analog input pin (A_{IN}) relative to V_{SS}	-0.6V to $V_{DD} + 0.6V$
SDA and SCL pins relative to V_{SS}	-0.6V to $V_{DD} + 1.0V$
Storage Temperature (T_{STG})	-65°C to +150°C
Ambient Temperature (T_A) with power applied.....	-65°C to +125°C
Maximum Junction Temperature (T_J).....	+150°C
ESD protection on all pins (HBM).....	≥ 4 kV

† **Notice:** Stresses above those listed under “Maximum Ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device at these, or any other conditions above those indicated in the operation listings of this specification, is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

1.2 Electrical Characteristics

TABLE 1-1: DC ELECTRICAL SPECIFICATIONS

Operating Conditions: Unless otherwise indicated, all parameters apply at $V_{DD} = 5.0V$, $V_{SS} = GND$, $R_{PU} = 2\text{ k}\Omega$ (Note 1), $T_A = -40^\circ\text{C}$ to $+125^\circ\text{C}$, I ² C Fast Mode Timing: $f_{SCL} = 400\text{ kHz}$.						
Parameter	Symbol	Min.	Typical	Max.	Units	Conditions
DC Accuracy						
Resolution	—	12			bits	
Integral Nonlinearity	INL	—	±0.75	±2	LSB	
Differential Nonlinearity	DNL	—	±0.5	±1	LSB	No missing codes
Offset Error	—	—	±0.75	±2	LSB	
Gain Error	—	—	-1	±3	LSB	
Dynamic Performance						
Total Harmonic Distortion	THD	—	-82	—	dB	$V_{IN} = 0.1V$ to $4.9V$ at $f_{IN} = 1\text{ kHz}$
Signal-to-Noise and Distortion	SINAD	—	72	—	dB	$V_{IN} = 0.1V$ to $4.9V$ at $f_{IN} = 1\text{ kHz}$
Spurious Free Dynamic Range	SFDR	—	86	—	dB	$V_{IN} = 0.1V$ to $4.9V$ at $f_{IN} = 1\text{ kHz}$
Analog Input						
Input Voltage Range	—	$V_{SS} - 0.3$	—	$V_{DD} + 0.3$	V	$2.7V \leq V_{DD} \leq 5.5V$
Leakage Current	—	-1	—	+1	μA	
Power Requirements						
Operating Voltage	V_{DD}	2.7	—	5.5	V	
Conversion Current	I_{DD}	—	175	250	μA	
Standby Current	I_{DDS}	—	0.005	1	μA	$-40^\circ\text{C} \leq T_A \leq +85^\circ\text{C}$ SDA/SCL = V_{DD} , Note 3
		—	0.9	—	μA	$+85^\circ\text{C} \leq T_A \leq +125^\circ\text{C}$ SDA/SCL = V_{DD} , Note 3
Active Bus Current	I_{DDA}	—	—	120	μA	SDA/SCL = V_{SS} to V_{DD} at $f_{SCL} = 400\text{ kHz}$

- Note 1:** Pull-up resistor on SDA and SCL lines.
Note 2: This parameter is periodically sampled and not 100% tested.
Note 3: Refer to [Figure 2-36](#) and [Figure 2-37](#) for more details.
Note 4: t_{ACQ} and t_{CONV} are dependent on internal oscillator timing. See [Figure 5-5](#) and [Figure 5-6](#) in regards to SCL.
Note 5: Sample time is the period between conversions after the address byte is sent to MCP3221. Refer to [Figure 5-6](#).

TABLE 1-1: DC ELECTRICAL SPECIFICATIONS (CONTINUED)

Operating Conditions: Unless otherwise indicated, all parameters apply at $V_{DD} = 5.0V$, $V_{SS} = GND$, $R_{PU} = 2\text{ k}\Omega$ (Note 1), $T_A = -40^\circ\text{C}$ to $+125^\circ\text{C}$, I ² C Fast Mode Timing: $f_{SCL} = 400\text{ kHz}$.						
Parameter	Symbol	Min.	Typical	Max.	Units	Conditions
SDA/SCL (open-drain output)						
Output Data Format	—	Straight Binary			—	
High-Level Input Voltage	V_{IH}	$0.7 \times V_{DD}$	—	—	V	
Low-Level Input Voltage	V_{IL}	—	—	$0.3 \times V_{DD}$	V	
Low-Level Output Voltage	V_{OL}	—	—	0.4	V	$I_{OL} = 3\text{ mA}$, $R_{PU} = 1.53\text{ k}\Omega$
Hysteresis of Schmitt Trigger Inputs	V_{HYST}	—	$0.05 \times V_{DD}$	—	V	$f_{SCL} = 400\text{ kHz}$ only
Input Leakage Current	I_{LI}	-1	—	+1	μA	$V_{IN} = V_{SS}$ to $0.9 \times V_{DD}$
Output Leakage Current	I_{LO}	-1	—	+1	μA	$V_{OUT} = V_{SS}$ to $0.9 \times V_{DD}$
Pin Capacitance (all inputs/ outputs)	C_{IN} , C_{OUT}	—	—	10	pF	$T_A = +25^\circ\text{C}$, $f = 1\text{ MHz}$, Note 2
Bus Capacitance	C_B	—	—	400	pF	SDA drive low, 0.4V
Conversion Rate						
Conversion Time	t_{CONV}	—	8.96	—	μs	Note 4
Analog Input Acquisition Time	t_{ACQ}	—	1.12	—	μs	Note 4
Sample Rate	f_{SAMP}	—	—	22.3	ksps	$f_{SCL} = 400\text{ kHz}$, Note 5

Note 1: Pull-up resistor on SDA and SCL lines.

2: This parameter is periodically sampled and not 100% tested.

3: Refer to [Figure 2-36](#) and [Figure 2-37](#) for more details.

4: t_{ACQ} and t_{CONV} are dependent on internal oscillator timing. See [Figure 5-5](#) and [Figure 5-6](#) in regards to SCL.

5: Sample time is the period between conversions after the address byte is sent to MCP3221. Refer to [Figure 5-6](#).

TABLE 1-2: TEMPERATURE SPECIFICATIONS

Operating Conditions: Unless otherwise indicated, all parameters apply at $V_{DD} = 5.0V$, $V_{SS} = GND$.						
Parameter	Symbol	Min.	Typical	Max.	Units	Conditions
Temperature Ranges						
Specified Temperature Range	T_A	-40	—	+125	$^\circ\text{C}$	
Operating Temperature Range	T_{OP}	-40	—	+125	$^\circ\text{C}$	
Storage Temperature Range	T_{STG}	-65	—	+150	$^\circ\text{C}$	
Thermal Package Resistances						
Thermal Resistance,	θ_{JA}	—	256	—	$^\circ\text{C/W}$	

TABLE 1-3: TIMING SPECIFICATIONS

Operating Conditions: Unless otherwise indicated, all parameters apply at $V_{DD} = 2.7V$ to $5.5V$, $V_{SS} = GND$, $T_A = -40^\circ\text{C}$ to $+125^\circ\text{C}$.						
Parameter	Symbol	Min.	Typical	Max.	Units	Conditions
I²C Standard Mode						
Clock Frequency	f_{SCL}	0	—	100	kHz	
Clock High Time	t_{HIGH}	4000	—	—	ns	
Clock Low Time	t_{LOW}	4700	—	—	ns	
SDA and SCL Rise Time	t_R	—	—	1000	ns	From V_{IL} to V_{IH} , Note 1
SDA and SCL Fall Time	t_F	—	—	300	ns	From V_{IL} to V_{IH} , Note 1

Note 1: This parameter is periodically sampled and not 100% tested.

2: Time period during which the bus must be free before a new I²C transmission can start.

TABLE 1-3: TIMING SPECIFICATIONS (CONTINUED)

Operating Conditions: Unless otherwise indicated, all parameters apply at $V_{DD} = 2.7V$ to $5.5V$, $V_{SS} = GND$, $T_A = -40^{\circ}C$ to $+125^{\circ}C$.						
Parameter	Symbol	Min.	Typical	Max.	Units	Conditions
START Condition Hold Time	$t_{HD:STA}$	4000	—	—	ns	
START Condition Setup Time	$t_{SU:STA}$	4700	—	—	ns	
Data Input Setup Time	$t_{SU:DAT}$	250	—	—	ns	
STOP Condition Setup Time	$t_{SU:STO}$	4000	—	—	ns	
STOP Condition Hold time	$t_{HD:STD}$	4000	—	—	ns	
Output Valid from Clock	t_{AA}	—	—	3500	ns	
Bus Free Time	t_{BUF}	4700	—	—	ns	Note 2
Input Filter Spike Suppression	t_{SP}	—	—	50	ns	SDA and SCL pins, Note 1
I²C Fast Mode						
Clock Frequency	f_{SCL}	0	—	400	kHz	
Clock High Time	t_{HIGH}	600	—	—	ns	
Clock Low Time	t_{LOW}	1300	—	—	ns	
SDA and SCL Rise Time	t_R	$20 + 0.1 \times C_B$	—	300	ns	From V_{IL} to V_{IH} , Note 1
SDA and SCL Fall Time	t_F	$20 + 0.1 \times C_B$	—	300	ns	From V_{IL} to V_{IH} , Note 1
START Condition Hold Time	$t_{HD:STA}$	600	—	—	ns	
START Condition Setup Time	$t_{SU:STA}$	600	—	—	ns	
Data Input Hold Time	$t_{HD:DAT}$	0	—	0.9	ms	
Data Input Setup Time	$t_{SU:DAT}$	100	—	—	ns	
STOP Condition Setup Time	$t_{SU:STO}$	600	—	—	ns	
STOP Condition Hold Time	$t_{HD:STD}$	600	—	—	ns	
Output Valid from Clock	t_{AA}	—	—	900	ns	
Bus Free Time	t_{BUF}	1300	—	—	ns	Note 2
Input Filter Spike Suppression	t_{SP}	—	—	50	ns	SDA and SCL pins, Note 1

Note 1: This parameter is periodically sampled and not 100% tested.

Note 2: Time period during which the bus must be free before a new I²C transmission can start.

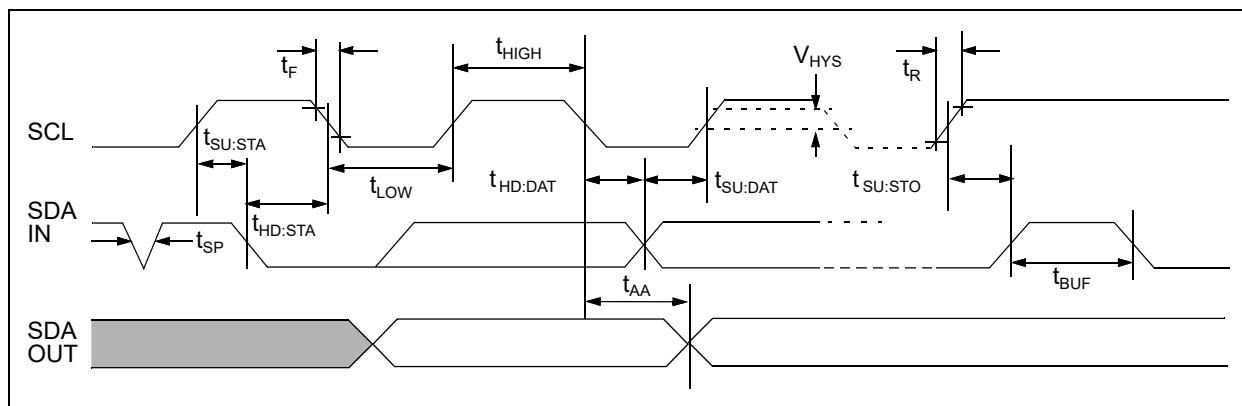


FIGURE 1-1: I²C Standard and Fast Mode Bus Timing Data.

2.0 TYPICAL PERFORMANCE CURVES

Note: The graphs and tables provided following this note are a statistical summary based on a limited number of samples and are provided for informational purposes only. The performance characteristics listed herein are not tested or guaranteed. In some graphs or tables, the data presented may be outside the specified operating range (for example, outside specified power supply range) and therefore outside the warranted range.

2.1 Performance Graphs

Note: Unless otherwise indicated, $V_{DD} = 5V$, $V_{SS} = 0V$, I^2C Fast Mode Timing ($f_{SCL} = 400$ kHz), Continuous Conversion Mode ($f_{SAMP} = 22.3$ kps), $T_A = +25^\circ C$.

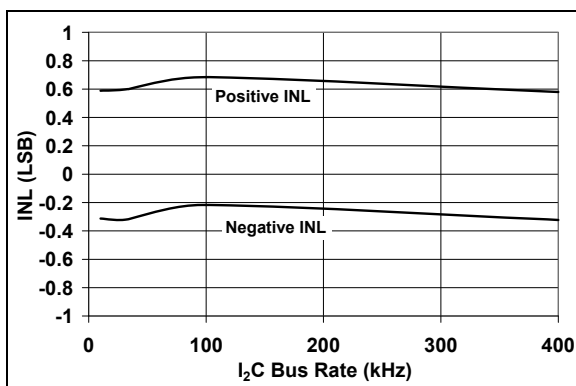


FIGURE 2-1: INL vs. Clock Rate.

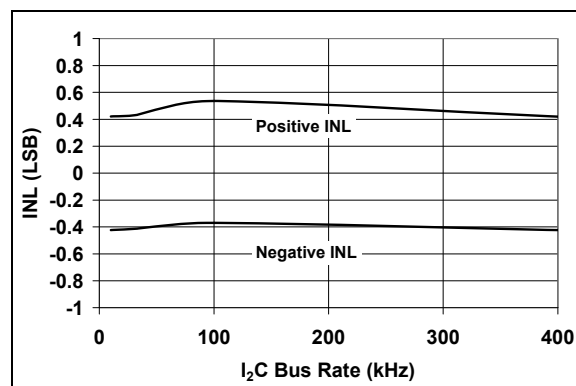


FIGURE 2-4: INL vs. Clock Rate, $V_{DD} = 2.7V$.

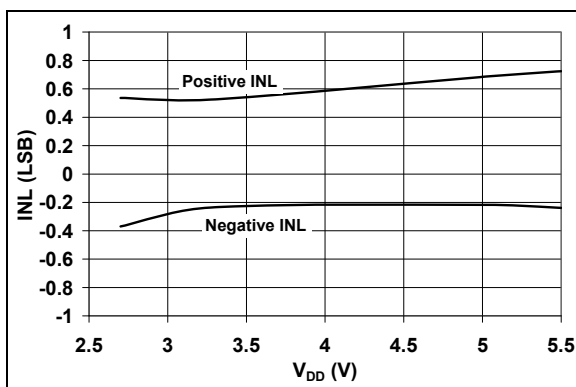


FIGURE 2-2: INL vs. V_{DD} . I^2C Standard Mode ($f_{SCL} = 100$ kHz).

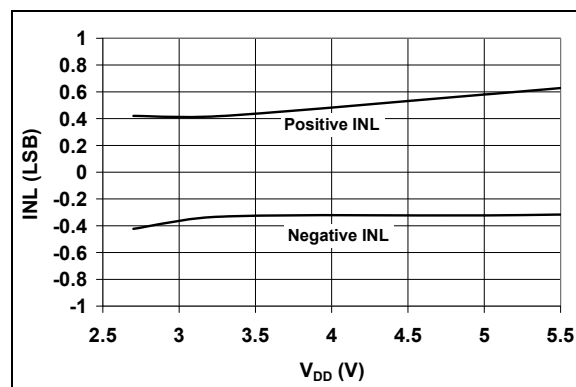


FIGURE 2-5: INL vs. V_{DD} . I^2C Fast Mode ($f_{SCL} = 400$ kHz).

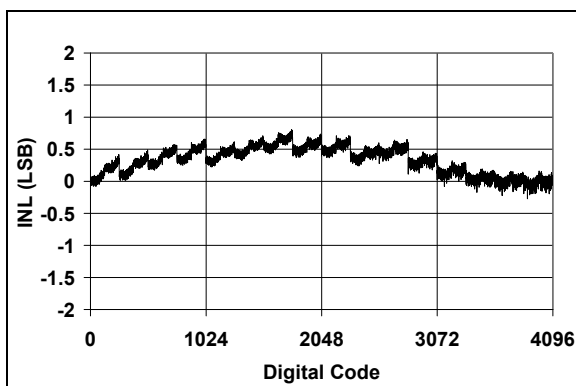


FIGURE 2-3: INL vs. Code. Representative Part.

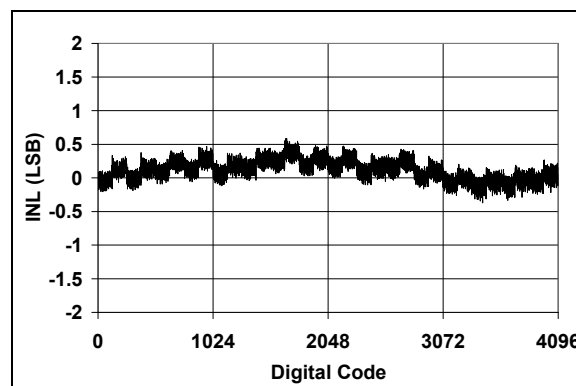


FIGURE 2-6: INL vs. Code. Representative Part, $V_{DD} = 2.7V$.

MCP3221

Note: Unless otherwise indicated, $V_{DD} = 5V$, $V_{SS} = 0V$, I^2C Fast Mode Timing ($f_{SCL} = 400$ kHz), Continuous Conversion Mode ($f_{SAMP} = 22.3$ ksp/s), $T_A = +25^\circ C$.

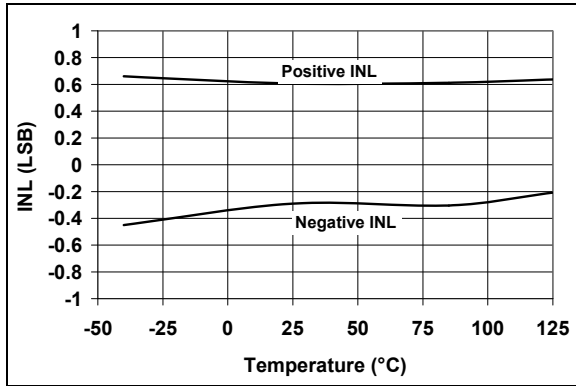


FIGURE 2-7: INL vs. Temperature.

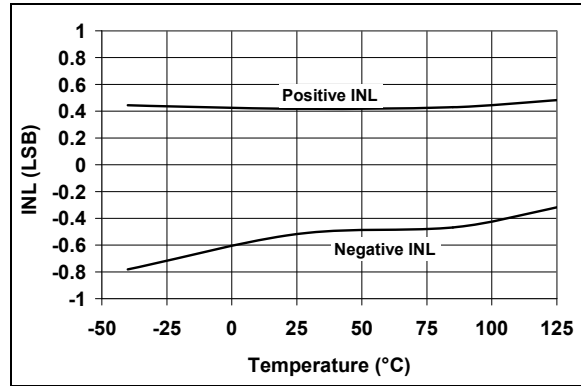


FIGURE 2-10: INL vs. Temperature, $V_{DD} = 2.7V$.

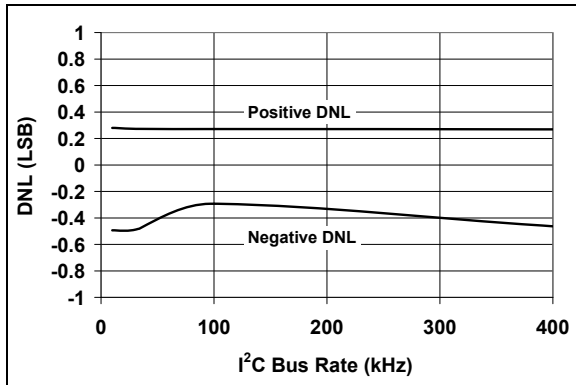


FIGURE 2-8: DNL vs. Clock Rate.

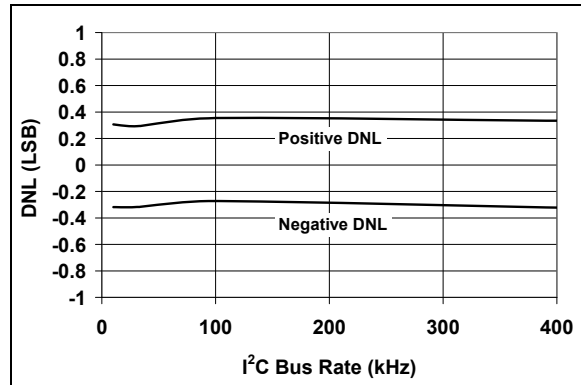


FIGURE 2-11: DNL vs. Clock Rate, $V_{DD} = 2.7V$.

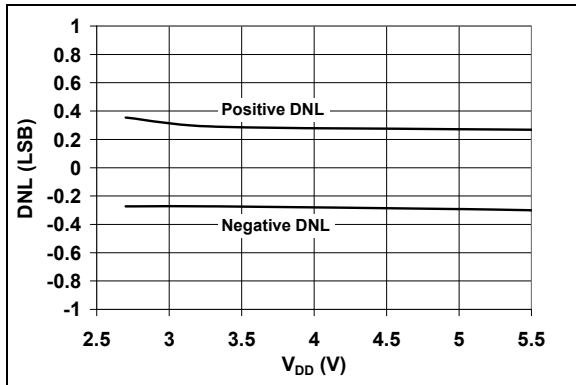


FIGURE 2-9: DNL vs. V_{DD} . I^2C Standard Mode ($f_{SCL} = 100$ kHz).

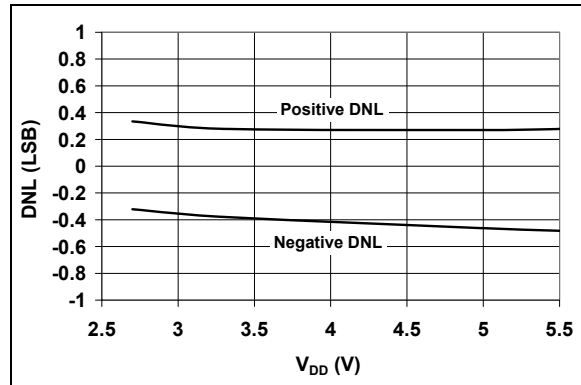


FIGURE 2-12: DNL vs. V_{DD} . I^2C Fast Mode ($f_{SCL} = 400$ kHz).

Note: Unless otherwise indicated, $V_{DD} = 5V$, $V_{SS} = 0V$, I²C Fast Mode Timing ($f_{SCL} = 400\text{ kHz}$), Continuous Conversion Mode ($f_{SAMP} = 22.3\text{ kSPS}$), $T_A = +25^\circ\text{C}$.

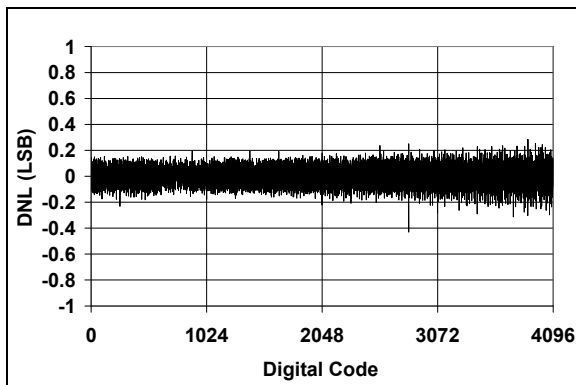


FIGURE 2-13: DNL vs. Code. Representative Part.

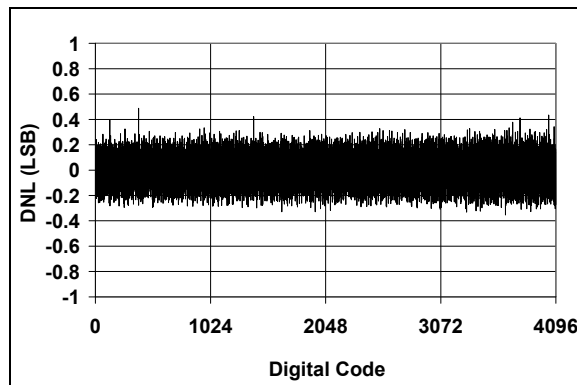


FIGURE 2-16: DNL vs. Code. Representative Part, $V_{DD} = 2.7V$.

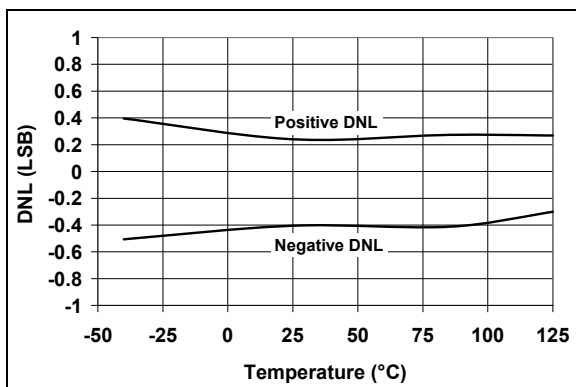


FIGURE 2-14: DNL vs. Temperature.

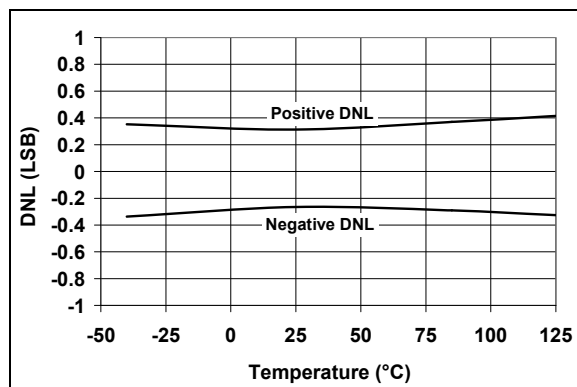


FIGURE 2-17: DNL vs. Temperature, $V_{DD} = 2.7V$.

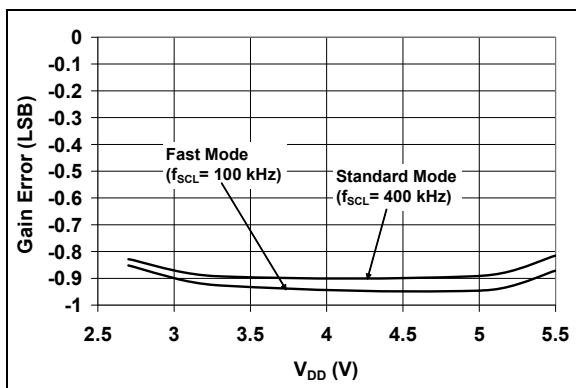


FIGURE 2-15: Gain Error vs. V_{DD} .

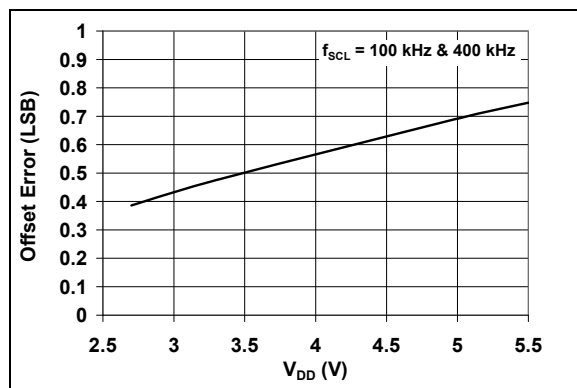


FIGURE 2-18: Offset Error vs. V_{DD} .

Note: Unless otherwise indicated, $V_{DD} = 5V$, $V_{SS} = 0V$, I²C Fast Mode Timing ($f_{SCL} = 400\text{ kHz}$), Continuous Conversion Mode ($f_{SAMP} = 22.3\text{ kSPS}$), $T_A = +25^\circ\text{C}$.

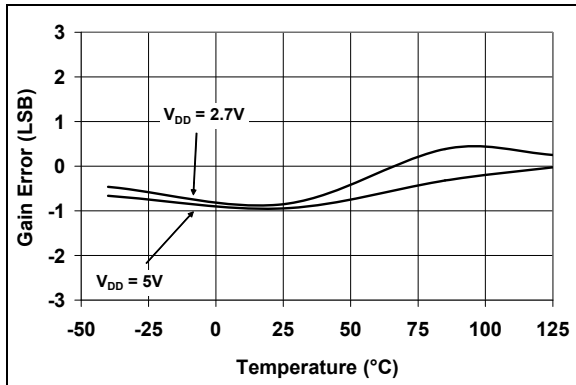


FIGURE 2-19: Gain Error vs. Temperature.

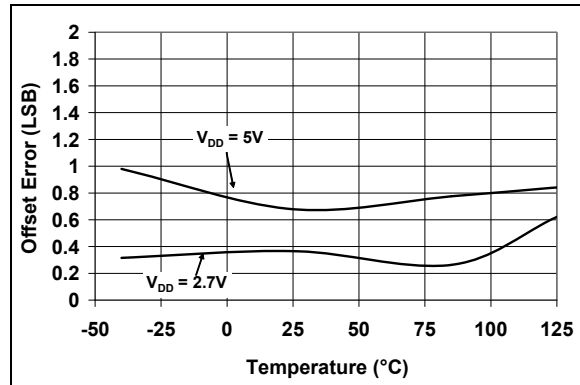


FIGURE 2-22: Offset Error vs. Temperature.

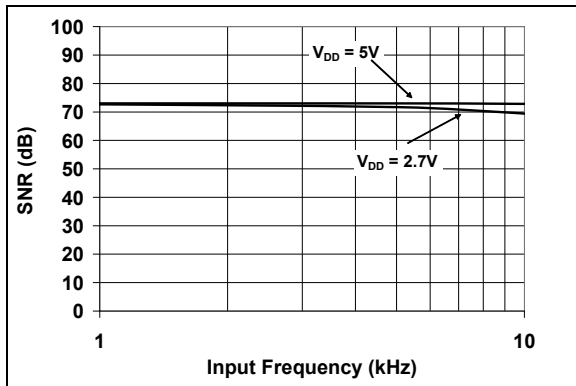


FIGURE 2-20: SNR vs. Input Frequency.

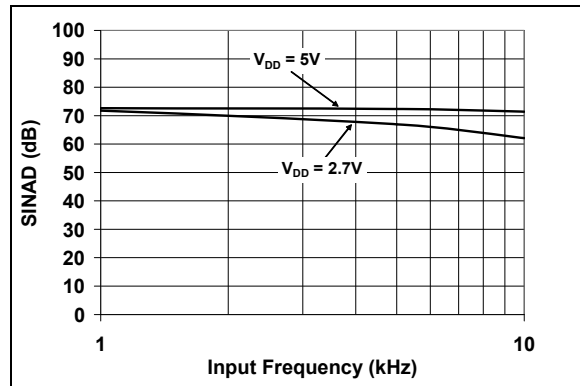


FIGURE 2-23: SINAD vs. Input Frequency.

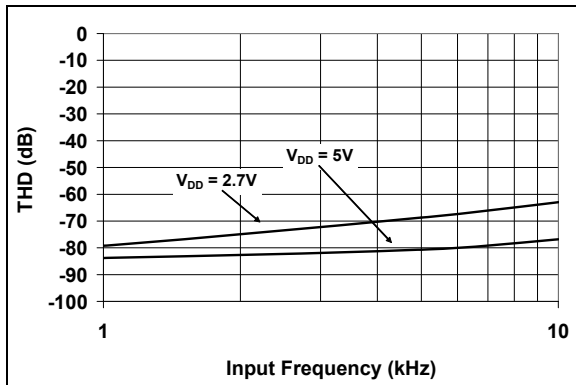


FIGURE 2-21: THD vs. Input Frequency.

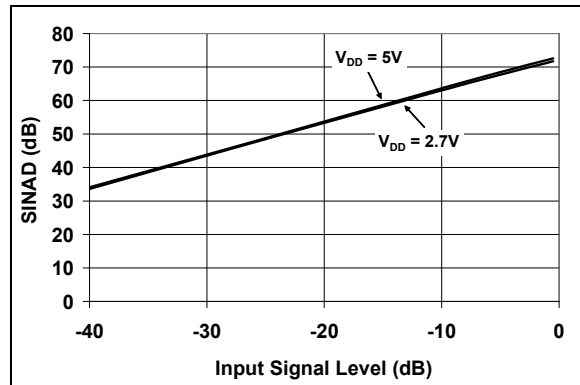


FIGURE 2-24: SINAD vs. Input Signal Level.

Note: Unless otherwise indicated, $V_{DD} = 5V$, $V_{SS} = 0V$, I^2C Fast Mode Timing ($f_{SCL} = 400$ kHz), Continuous Conversion Mode ($f_{SAMP} = 22.3$ ksp/s), $T_A = +25^\circ C$.

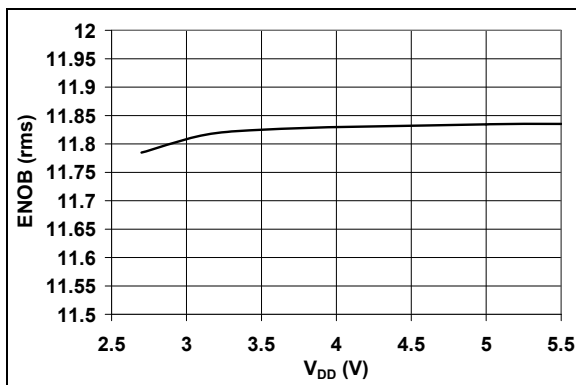


FIGURE 2-25: $ENOB$ vs. V_{DD} .

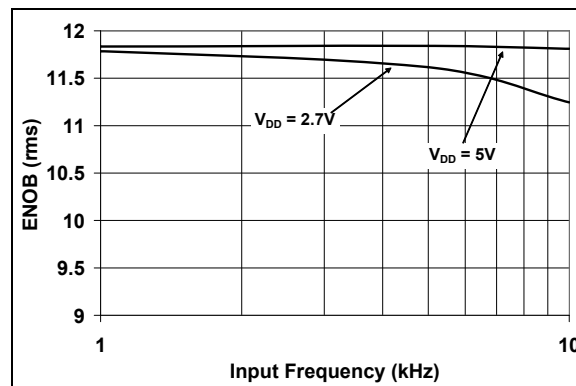


FIGURE 2-28: $ENOB$ vs. Input Frequency.

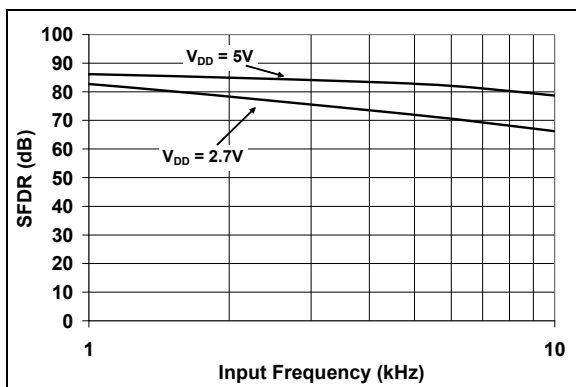


FIGURE 2-26: $SFDR$ vs. Input Frequency.

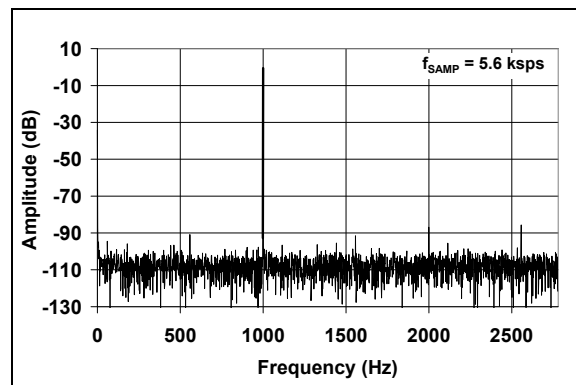


FIGURE 2-29: Spectrum Using I^2C Standard Mode. Representative Part, 1 kHz Input Frequency.

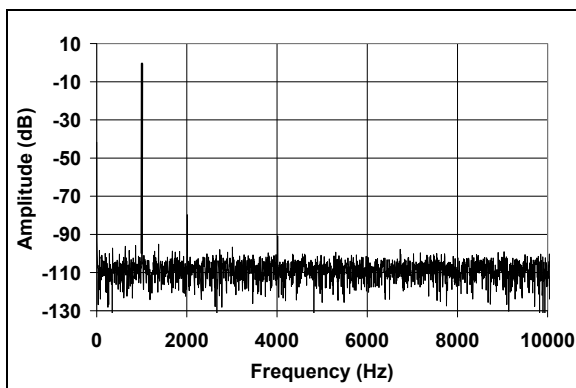


FIGURE 2-27: Spectrum Using I^2C Fast Mode. Representative Part, 1 kHz Input Frequency.

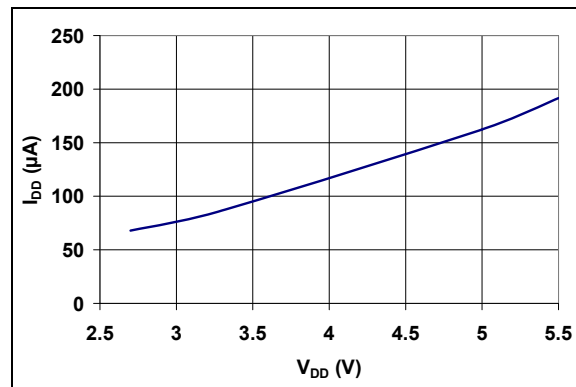


FIGURE 2-30: I_{DD} (Conversion) vs. V_{DD} .

MCP3221

Note: Unless otherwise indicated, $V_{DD} = 5V$, $V_{SS} = 0V$, I²C Fast Mode Timing ($f_{SCL} = 400\text{ kHz}$), Continuous Conversion Mode ($f_{SAMP} = 22.3\text{ kSPS}$), $T_A = +25^\circ\text{C}$.

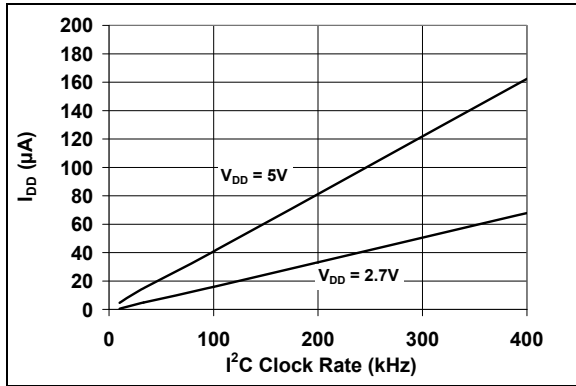


FIGURE 2-31: I_{DD} (Conversion) vs. Clock Rate.

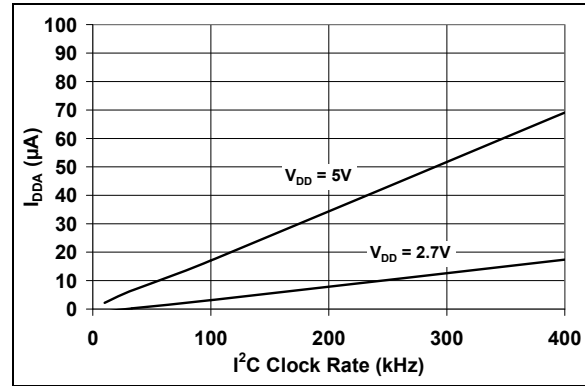


FIGURE 2-34: I_{DDA} (Active Bus) vs. Clock Rate.

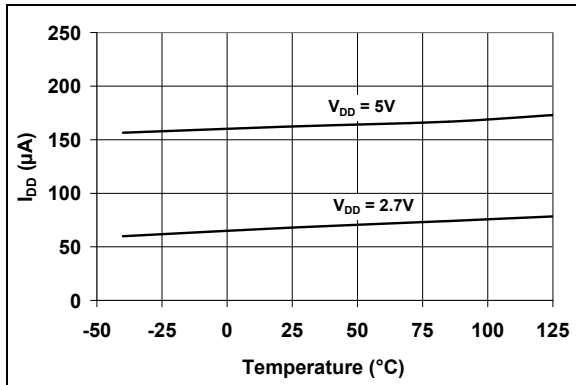


FIGURE 2-32: I_{DD} (Conversion) vs. Temperature.

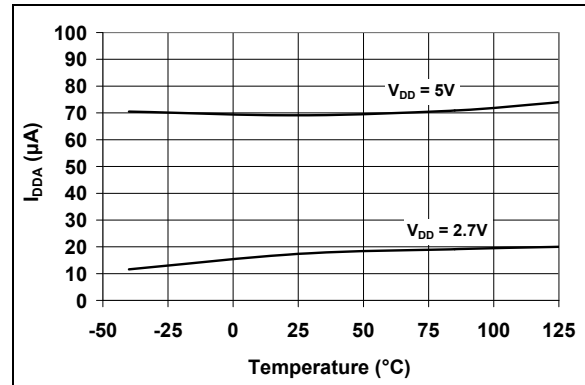


FIGURE 2-35: I_{DDA} (Active Bus) vs. Temperature.

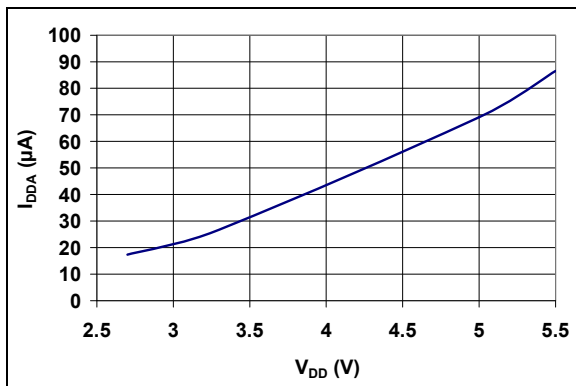


FIGURE 2-33: I_{DDA} (Active Bus) vs. V_{DD} .

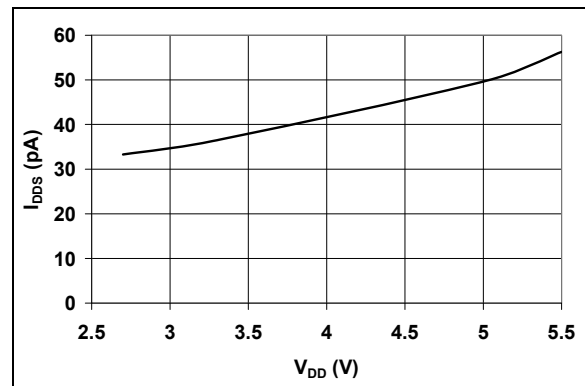


FIGURE 2-36: I_{DDS} (Standby) vs. V_{DD} .

Note: Unless otherwise indicated, $V_{DD} = 5V$, $V_{SS} = 0V$, I²C Fast Mode Timing ($f_{SCL} = 400\text{ kHz}$), Continuous Conversion Mode ($f_{SAMP} = 22.3\text{ ksps}$), $T_A = +25^\circ\text{C}$.

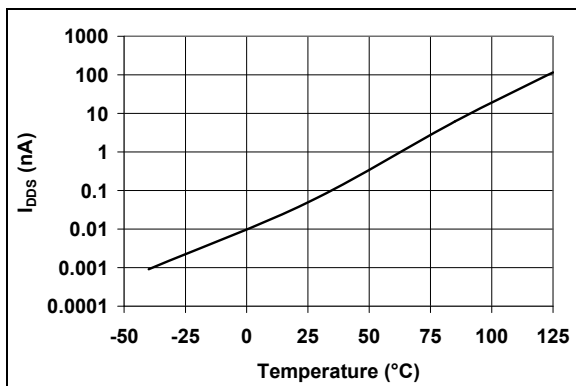


FIGURE 2-37: I_{DDS} (Standby) vs. Temperature.

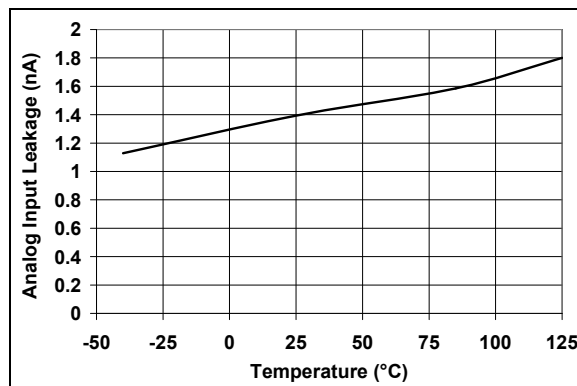


FIGURE 2-38: Analog Input Leakage vs. Temperature.

2.2 Test Circuits

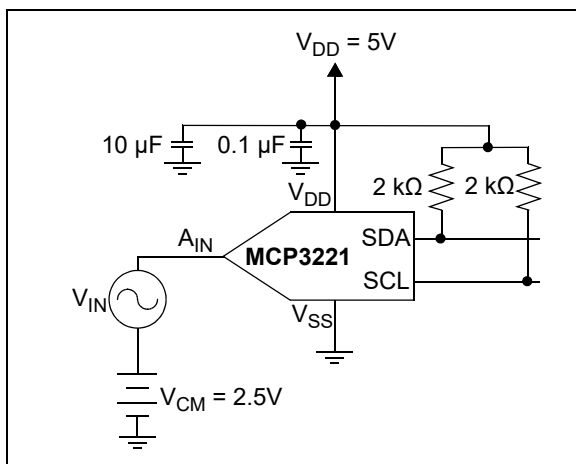


FIGURE 2-39: Typical Test Configuration.

MCP3221

3.0 PIN DESCRIPTION

Table 3-1 lists the function of the MCP3221 pins.

TABLE 3-1: MCP3221 PIN DESCRIPTIONS

Pin	Symbol	Description
1	V_{DD}	+2.7V to +5.5V Power Supply
2	V_{SS}	Ground
3	A_{IN}	Analog Input
4	SDA	Serial Data In/Out
5	SCL	Serial Clock In

3.1 V_{DD} and V_{SS}

The V_{DD} pin, with respect to V_{SS} , provides MCP3221 with power and also a voltage reference for the conversion process. Refer to [Section 6.4, "Device Power and Layout Considerations"](#) for tips on power and grounding.

3.2 Analog Input (A_{IN})

The A_{IN} pin is the input for the sample and hold circuitry of the Successive Approximation Register (SAR) converter. Ensure proper care when driving this pin. Refer to [Section 6.1, "Driving the Analog Input"](#) for more information. For proper conversions, the voltage on the A_{IN} pin can vary from V_{SS} to V_{DD} .

3.3 Serial Data (SDA)

SDA is a bidirectional pin that transfers addresses and data in and out of MCP3221. Being an open-drain terminal, the SDA bus requires a pull-up resistor connected to V_{DD} (typically 10 k Ω for $f_{SCL} = 100$ kHz and 2 k Ω for $f_{SCL} = 400$ kHz). Refer to [Section 6.2, "Connecting to the I2C Bus"](#) for more information.

For normal data transfer, the SDA pin is allowed to change only when the SCL pin is low. Changes when the SCL pin is high are reserved for indicating START and STOP conditions. Refer to [Section 5.1, "I2C Bus Characteristics"](#) for more information.

3.4 Serial Clock (SCL)

SCL is an input pin that synchronizes data transfers to and from MCP3221 on the SDA pin. As an open-drain terminal, the SCL bus requires a pull-up resistor connected to V_{DD} (typically 10 k Ω for $f_{SCL} = 100$ kHz and 2 k Ω for $f_{SCL} = 400$ kHz). Refer to [Section 6.2, "Connecting to the I2C Bus"](#) for more information.

For normal data transfer, the SDA pin is allowed to change only when the SCL pin is low. Changes when the SCL pin is high are reserved for indicating START and STOP conditions. Refer to [Section 6.1, "Driving the Analog Input"](#) for more information.

4.0 MCP3221 DEVICE OPERATION

4.1 General Description

MCP3221 employs a classic SAR architecture with an internal sample and hold capacitor to store the analog input during conversion. When acquisition time ends, the input switch of the converter opens and MCP3221 uses the collected charge on the internal sample and hold capacitor to produce a serial 12-bit digital output code. Acquisition time and conversion are self-timed using an internal clock. After each conversion, results are stored in a 12-bit register that can be read at any time.

Communication with MCP3221 is done over a 2-wire I²C interface. Maximum sample rates of 22.3 ksp/s are possible when the device is in a continuous-conversion mode and uses a SCL clock rate (f_{SCL}) of 400 kHz.

4.2 Digital Output Code

The digital output code produced by MCP3221 is a function of the input signal and power supply voltage, V_{DD} . As V_{DD} level decreases, LSB size decreases accordingly. Equation 4-1 shows the theoretical LSB size.

EQUATION 4-1:

$$\text{LSB Size} = \frac{V_{DD}}{4096}$$

Where:

LSB Size = Size of the Least Significant Bit (V)

V_{DD} = Power Supply Voltage (V)

The output code of MCP3221 is transmitted serially with the MSB first. The format of the code is straight binary.

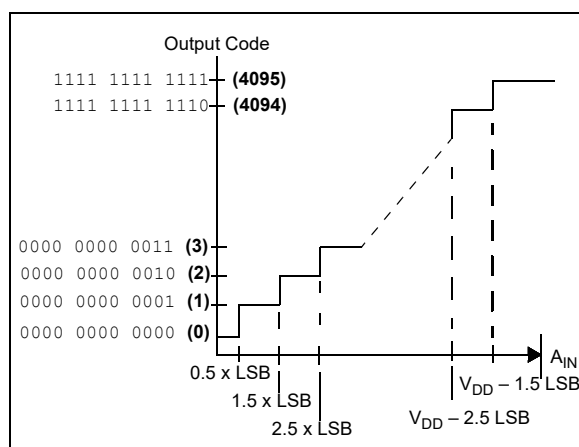


FIGURE 4-1: Transfer Function.

4.3 Conversion Time (t_{CONV})

Conversion time is the duration needed to obtain the digital result after the analog input is disconnected from the holding capacitor.

For MCP3221, the conversion time is typically 8.96 μ s. This period depends only on the internal oscillator and is not influenced of SCL.

4.4 Acquisition Time (t_{ACQ})

Acquisition time is the period needed to charge the sampling capacitor array.

For MCP3221, the acquisition time is typically 1.12 μ s. This period depends only on the internal oscillator and is not influenced of SCL.

4.5 Sample Rate

Sample rate is the reciprocal of the maximum period between the acquisition of the first conversion to the acquisition of the second conversion.

Sample rate can be measured using either single or continuous conversions.

A single conversion includes the following:

- A single START bit
- A single Address byte
- Two Data bytes
- A single STOP bit

For single conversions, the sample rate is measured between consecutive START bits.

For continuous conversions, the sample rate is measured between consecutive conversions or for a total of 18 clocks (two data bytes and two Acknowledge bits). The Host device requests a continuous conversion by issuing an Acknowledge after a conversion. Refer to Section 5.2, "Device Addressing" for more information.

4.6 Differential Nonlinearity (DNL)

In the ideal ADC transfer function, each digital code has a uniform width, meaning the difference in analog input voltage is constant from one code transition point to the next. Differential nonlinearity (DNL) measures the deviation of any code in the transfer function from an ideal code width of 1 LSB. The DNL is calculated by subtracting the positions of consecutive code transition points after compensating for any gain and offset errors. A positive DNL indicates that a code is longer than the ideal code width, while a negative DNL means a code is shorter than the ideal width.

4.7 Integral Nonlinearity (INL)

Integral nonlinearity (INL) is the result of cumulative DNL errors and it measures the deviation of the overall transfer function from a linear response. The MCP3221 ADC uses the end-point measurement method to determine INL.

4.8 Offset Error

The offset error is the deviation of the code transition points that occurs across all output codes, effectively shifting the entire ADC transfer function. Measure this error by determining the difference between the actual position of the first code transition and its ideal position. Ideally, the first code transition occurs at 0.5 LSB above V_{SS} .

4.9 Gain Error

The gain error is the deviation from the ideal slope of the ADC transfer function. Before determining the gain error, first measure the offset error and subtract it from the conversion result. Then calculate the gain error by determining the difference between the actual position of the last code transition and its ideal position. Ideally, the last code transition occurs at 1.5 LSB below Full Scale or V_{DD} .

4.10 Conversion Current (I_{DD})

The conversion current, I_{DD} , is the amount of current consumed by MCP3221 averaged over the time needed to perform a 12-bit conversion.

4.11 Active Bus Current (I_{DDA})

The active bus current, I_{DDA} , is the amount of current consumed by MCP3221 averaged over the time needed to monitor the I²C bus. This is the current consumed by the device while it is not addressed.

4.12 Standby Current (I_{DDs})

The standby current, I_{DDs} , is the amount of current consumed by MCP3221 averaged over the time when the SCL and SDA lines are inactive (no conversions occur and no data are outputted).

4.13 I²C Timing

MCP3221 uses the following two I²C timing modes as defined per I²C specifications:

- Standard Mode — the SCL frequency is 100 kHz:
 $f_{SCL} = 100 \text{ kHz}$.
- Fast Mode — the SCL frequency is 400 kHz:
 $f_{SCL} = 400 \text{ kHz}$.

5.0 SERIAL COMMUNICATIONS

5.1 I²C Bus Characteristics

The following bus protocols are defined:

- Data transfer can be initiated only when the bus is not busy.
- During data transfer, the data line must remain stable whenever the clock (SCL) line is high. Any changes in the data line while the SCL line is high are interpreted as a START or STOP condition.

Accordingly, the following bus conditions are defined (refer to [Figure 5-1](#)).

5.1.1 BUS NOT BUSY (A)

Both data and clock lines remain high.

5.1.2 START DATA TRANSFER (B)

A high-to-low transition on the data (SDA) line while SCL is high determines a START condition.

Note: All commands must begin with a START condition.

5.1.3 STOP DATA TRANSFER (C)

A low-to-high transition on the SDA line while SCL is high determines a STOP condition.

Note: All commands must end with a STOP condition.

5.1.4 DATA VALID (D)

Valid data is transmitted on the SDA line when, after a START condition, the data line is stable for the duration of the SCL high period.

The data on the line must change only during the SCL low period. There is one clock pulse per bit of data.

Every data transfer begins with a START condition and terminates with a STOP condition. The Host device determines the number of data bytes transferred between the START and STOP conditions. This number is unlimited.

5.1.5 ACKNOWLEDGE

Every receiving device, when addressed, generates an Acknowledge (ACK) bit after receiving each byte. The Host device must generate an extra clock pulse that is associated with this ACK bit.

The acknowledging device has to pull down the SDA line during the ACK clock pulse so that the SDA line is stable and low during the high period of the ACK clock pulse. Consider setup and hold times. During reads, the Host device must signal an end of data to the Client device by not generating an ACK bit when the last byte is clocked out of the client device. In this case, the Client device (MCP3221) releases the bus to allow the Host device to generate the STOP condition.

MCP3221 supports a bidirectional, 2-wire bus and data transmission protocol. The device that sends data onto the bus is called the transmitter and the device that receives the data is called the receiver. A Host device controls the bus. It produces the serial clock (SCL), manages bus access and generates START and STOP conditions. MCP3221 functions as a Client device. While both Host and Client devices can operate as transmitter or receiver, only the Host device determines what mode is active.

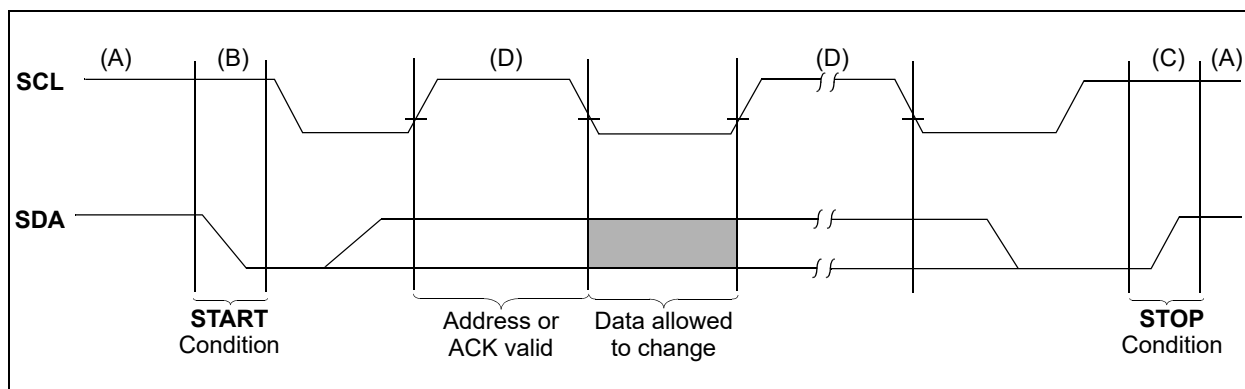


FIGURE 5-1: Data Transfer Sequence on the Serial Bus.

5.2 Device Addressing

The address byte is the first byte received after the START condition from the Host device. The first part of the address byte consists of a 4-bit device code. For MCP3221, this code is set to 1001.

The device code is followed by three address bits: A2, A1 and A0. The default address bits are 101.

Note: Contact the Microchip factory for additional address bit options.

The address bits allow up to eight MCP3221 devices on the same bus and they specify the target device.

The 8th and last bit of the address byte is called the Read/Write (R/W) bit. It determines the operation type requested by the Host device:

- To read conversion data, the R/W bit is set to 1.
- To write to a registry, the R/W bit is set to 0.

However, MCP3221 has no writable registers. Thus, if the R/W address bit received by MCP3221 is set to 1, it triggers a conversion.

MCP3221 is a Client device that is compatible with the 2-wire I²C serial interface protocol. Figure 6-2 shows a hardware connection diagram. The microcontroller unit (MCU) on the Host device initiates communication and sends a START condition followed by the address byte.

After MCP3221 completes the conversion(s), the MCU must send a STOP condition to end communication.

When receiving R/W = 0, MCP3221 returns an ACK and then releases the bus. This behavior can be used to poll the device. For more information, refer to Section 6.3, "Device Polling".

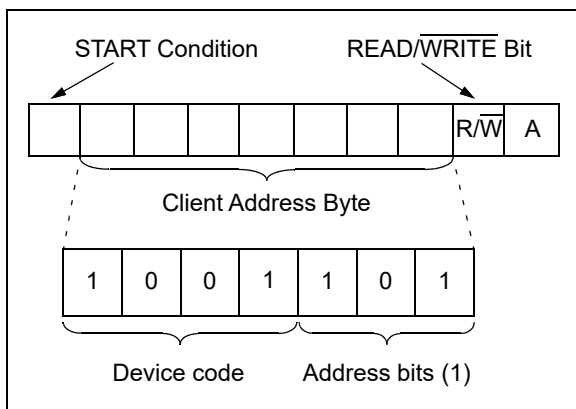


FIGURE 5-2: Device Addressing.

5.3 Executing a Conversion

This section describes the details of communicating with MCP3221. These include the initiation of sample and hold acquisition, the reading the conversion data and the execution of multiple conversions.

5.3.1 INITIATE SAMPLE AND HOLD ACQUISITION

The acquisition and conversion of the input signal begins with the falling edge of the R/W bit, part of the address byte. At this point, the internal clock initiates the sample, hold and conversion cycle, all of them being internal to the ADC.

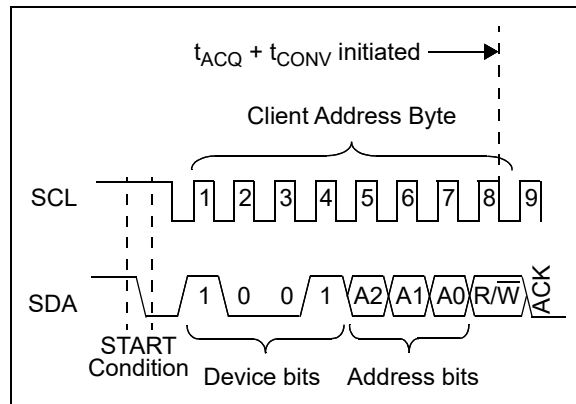


FIGURE 5-3: Conversion Initiation, Address Byte.

The input signal is initially sampled with the first SCL falling edge following the transmission of a logic-high R/W bit. Additionally, with the SCL rising edge, the ADC transmits an ACK = 0 bit. The Host device releases the data bus during this clock pulse to allow MCP3221 to pull the line low (see Figure 5-3).

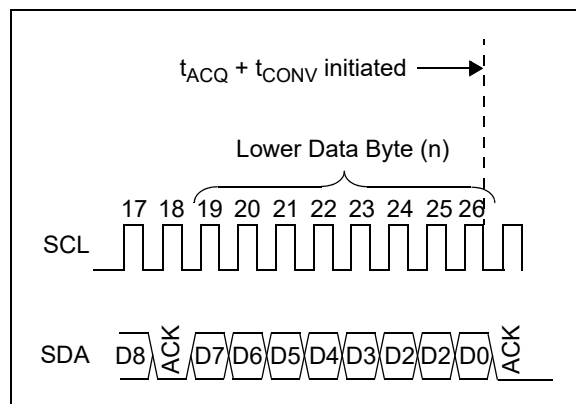


FIGURE 5-4: Conversion Initiation, Continuous Conversions.

5.3.2 READING CONVERSION DATA

After MCP3221 acknowledges the address byte, the device transmits four 0 bits, followed by the upper four data bits of the conversion. The Host device responds to this byte with an ACK = Low. With the following eight clock pulses, MCP3221 transmits the lower eight data bits of the conversion. After receiving these eight bits, the Host device replies with an ACK = high, indicating to the MCP3221 device that it request no further data. The Host device can send a STOP condition to terminate the transmission.

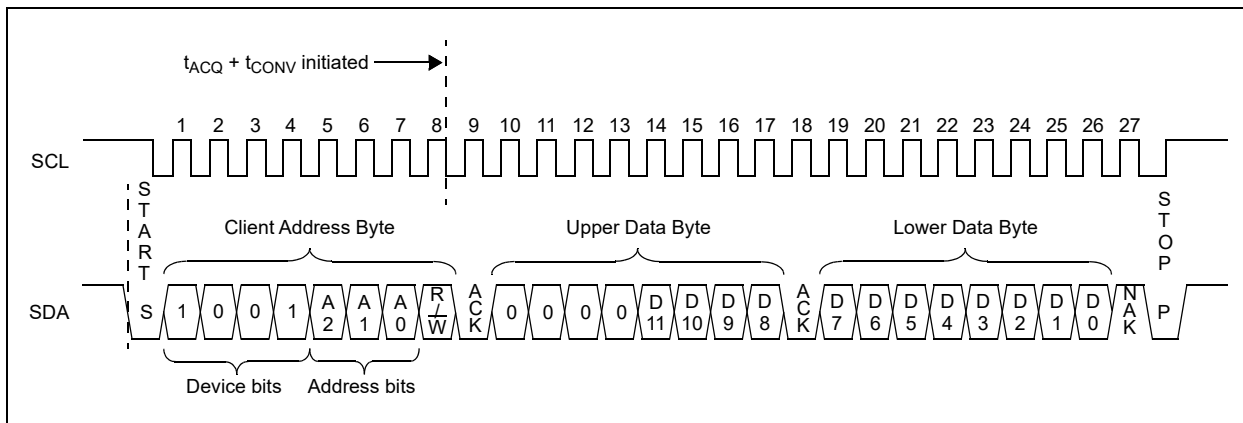


FIGURE 5-5: Executing a Conversion.

5.3.3 CONSECUTIVE CONVERSIONS

For consecutive conversions, a new sampling starts on the falling edge of the LSB of the conversion result. The device provides the 12-bit conversion results in a two-byte format. Figure 5-6 shows the timing diagram.

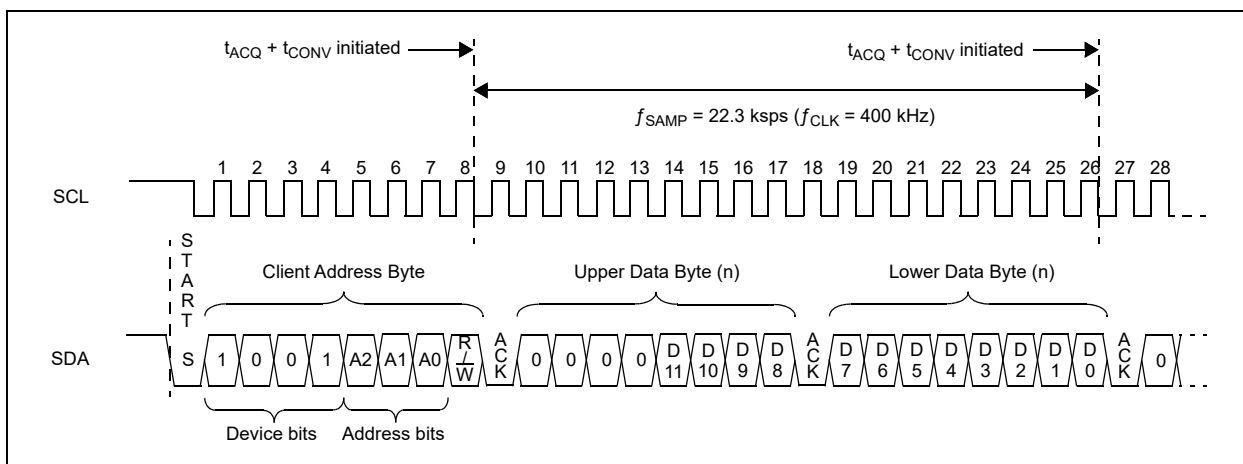


FIGURE 5-6: Continuous Conversion.

6.0 APPLICATION INFORMATION

6.1 Driving the Analog Input

MCP3221 has a single-ended analog input, A_{IN} . For proper conversion results, ensure the voltage at the A_{IN} pin remains between V_{SS} and V_{DD} . If the converter has no offset error, gain error, INL or DNL errors and the voltage level of A_{IN} is $\leq V_{SS} + 0.5 \times \text{LSB}$, the output code is 000h. If the voltage at A_{IN} is $\geq V_{DD} - 1.5 \times \text{LSB}$, then the output code is instead FFFh.

The analog input model in Figure 6-1 shows how the source impedance (R_S) adds to the impedance of the internal sampling switch (R_{SS}), directly affecting the time required to charge capacitor C_{SAMPLE} . Thus, a larger source impedance increases the offset, gain and INL errors of the conversion. Ideally, $R_S \approx 0\Omega$. This is done using an operational amplifier such as MCP6022, which has a closed-loop output impedance of tens of ohms.

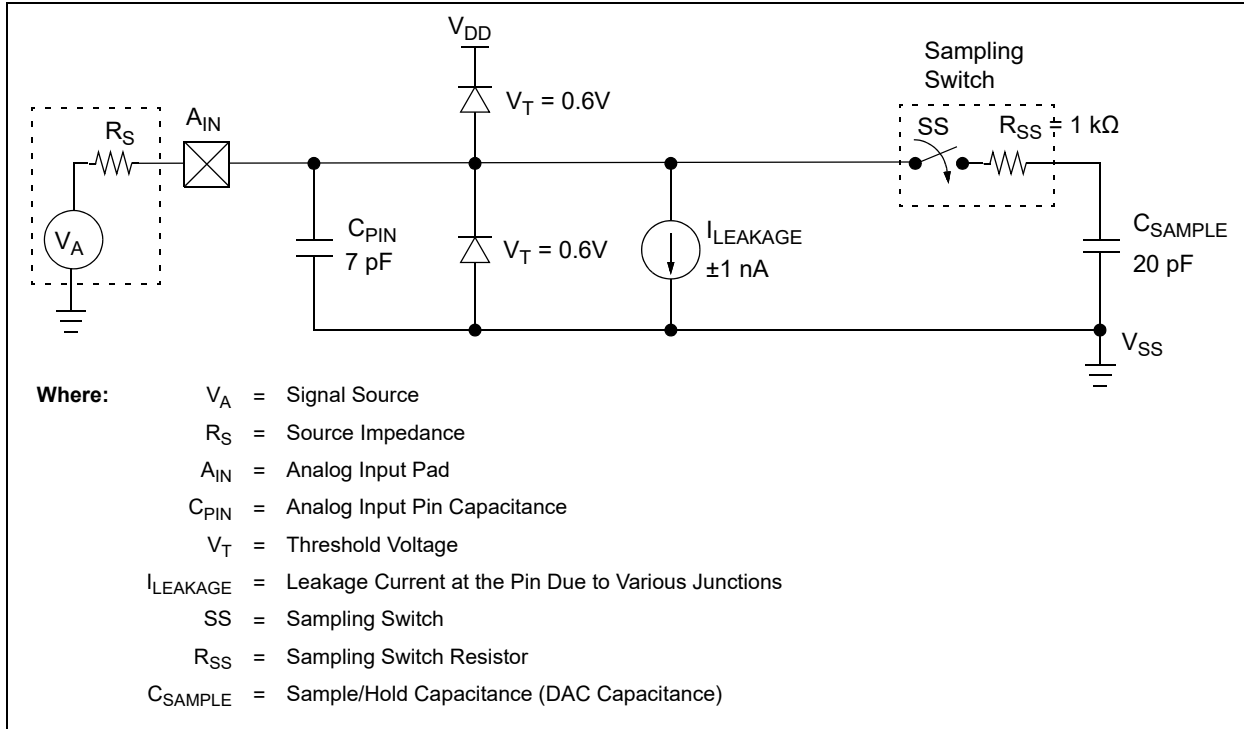


FIGURE 6-1: Analog Input Model, A_{IN} .

6.2 Connecting to the I²C Bus

The I²C bus is an open collector bus, requiring pull-up resistors (R_{PU}) connected to the SDA and SCL lines. Figure 6-2 shows this configuration.

The number of devices connected to the bus is limited only by the maximum bus capacitance of 400 pF. Figure 6-3 shows a possible configuration with multiple devices connected to the same I²C bus.

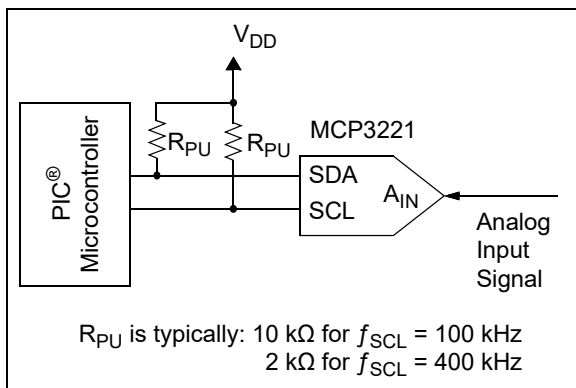


FIGURE 6-2: Pull-up Resistors on I²C Bus.

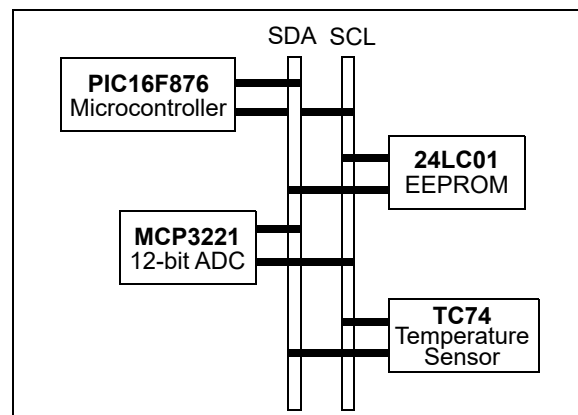


FIGURE 6-3: Multiple Devices on I²C Bus.

6.3 Device Polling

Sometimes, it is necessary to test for the presence of MCP3221 on the I²C bus without doing a conversion. This is described in Figure 6-4, where the R/W bit in the address byte is set to 0. MCP3221 acknowledges by pulling the SDA line low during the ACK clock and then releases the I²C bus. The Host device can issue a STOP or a repeated START condition to continue I²C communication.

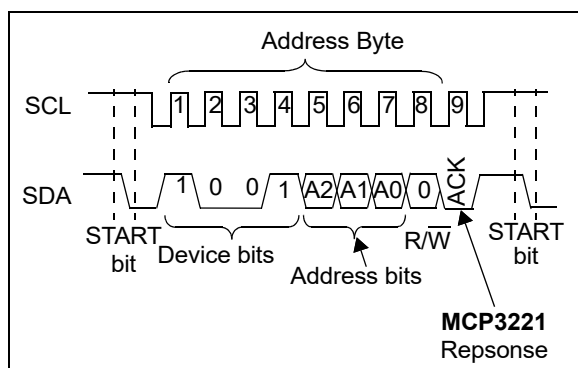


FIGURE 6-4: Device Polling.

6.4 Device Power and Layout Considerations

6.4.1 POWERING MCP3221

V_{DD} supplies MCP3221 with power and a reference voltage. Microchip recommends connecting a 0.1 μ F bypass capacitor to V_{DD} in parallel with a 10 μ F capacitor to attenuate higher frequency noise present in certain systems. Figure 6-5 shows this configuration.

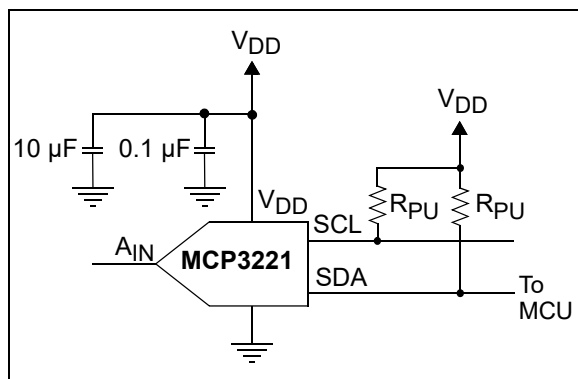


FIGURE 6-5: Powering MCP3221.

Note: When power-down of the MCP3221 is needed during applications (after power-up), it is highly recommended to bring down the V_{DD} to V_{SS} level. This can guarantee a Full Reset of the device for the next power-up cycle.

6.5 Layout Considerations

When laying out a printed circuit board for use with analog components, care should be taken to reduce noise wherever possible. A bypass capacitor from V_{DD} to ground must be used always with this device and placed as close as possible to the device pin. A bypass capacitor value of 0.1 μ F is recommended.

Digital and analog traces should be separated as much as possible on the board, with no traces running underneath the device or the bypass capacitor. Extra precautions should be taken to keep traces with high-frequency signals (such as clock lines) as far as possible from analog traces.

Use of an analog ground plane is recommended in order to keep the ground potential the same for all devices on the board. Providing V_{DD} connections to devices in a Star configuration can also reduce noise by eliminating current return paths and associated errors (Figure 6-7). For more information on layout tips when using the MCP3221 or other ADC devices, refer to the Microchip Technology Application Note, "AN688 Layout Tips for 12-Bit A/D Converter Application" (DS00688).

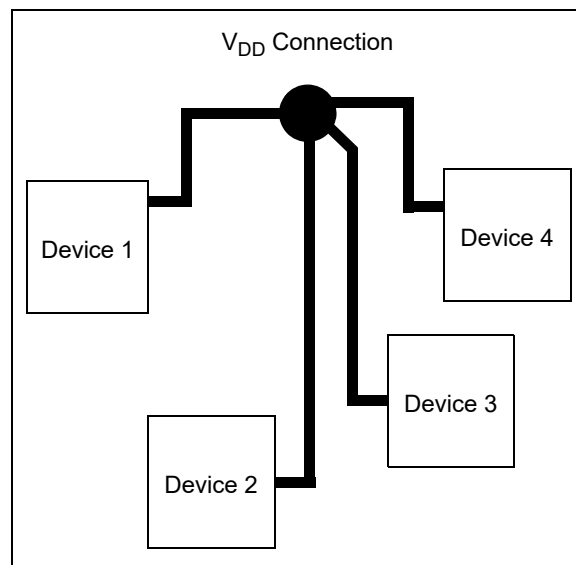


FIGURE 6-6: V_{DD} Traces Arranged in a Star Configuration to Reduce Errors Caused by Current Return Paths.

6.5.1 USING A REFERENCE FOR THE SUPPLY VOLTAGE

MCP3221 uses V_{DD} as power and also as a reference voltage. In certain applications, it is necessary to use a stable reference to achieve the required accuracy. Figure 6-7 illustrates an example of using the MCP1541 voltage reference device to provide a stable supply voltage.

MCP3221

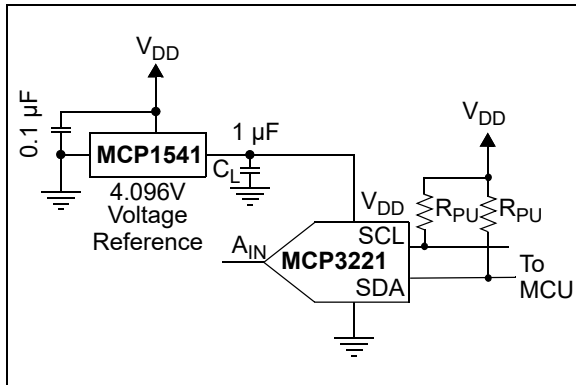
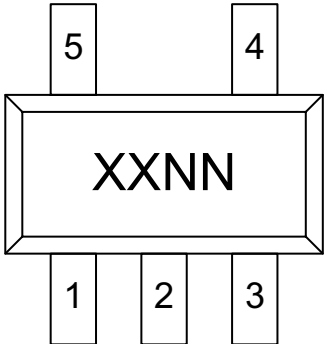


FIGURE 6-7: *Stable Power and Reference Configuration.*

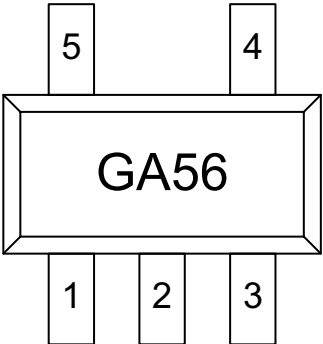
7.0 PACKAGE INFORMATION

7.1 Package Marking Information

MCP3221
5-Lead SOT-23



Example:



Part Number	Address Option	Package Marking XX SOT-23
MCP3221A0T-E/OT	000	GE
MCP3221A1T-E/OT	001	GH
MCP3221A2T-E/OT	010	GB
MCP3221A3T-E/OT	011	GC
MCP3221A4T-E/OT	100	GD
MCP3221A5T-E/OT	101	GA ^(*)
MCP3221A6T-E/OT	110	GF
MCP3221A7T-E/OT	111	GG

* Default option. Contact Microchip Factory for other address options.

Legend:

XX...X Customer-specific information

Y Year code (last digit of calendar year)

YY Year code (last 2 digits of calendar year)

WW Week code (week of January 1 is week '01')

NNN Alphanumeric traceability code

 Pb-free JEDEC designator for Matte Tin (Sn)

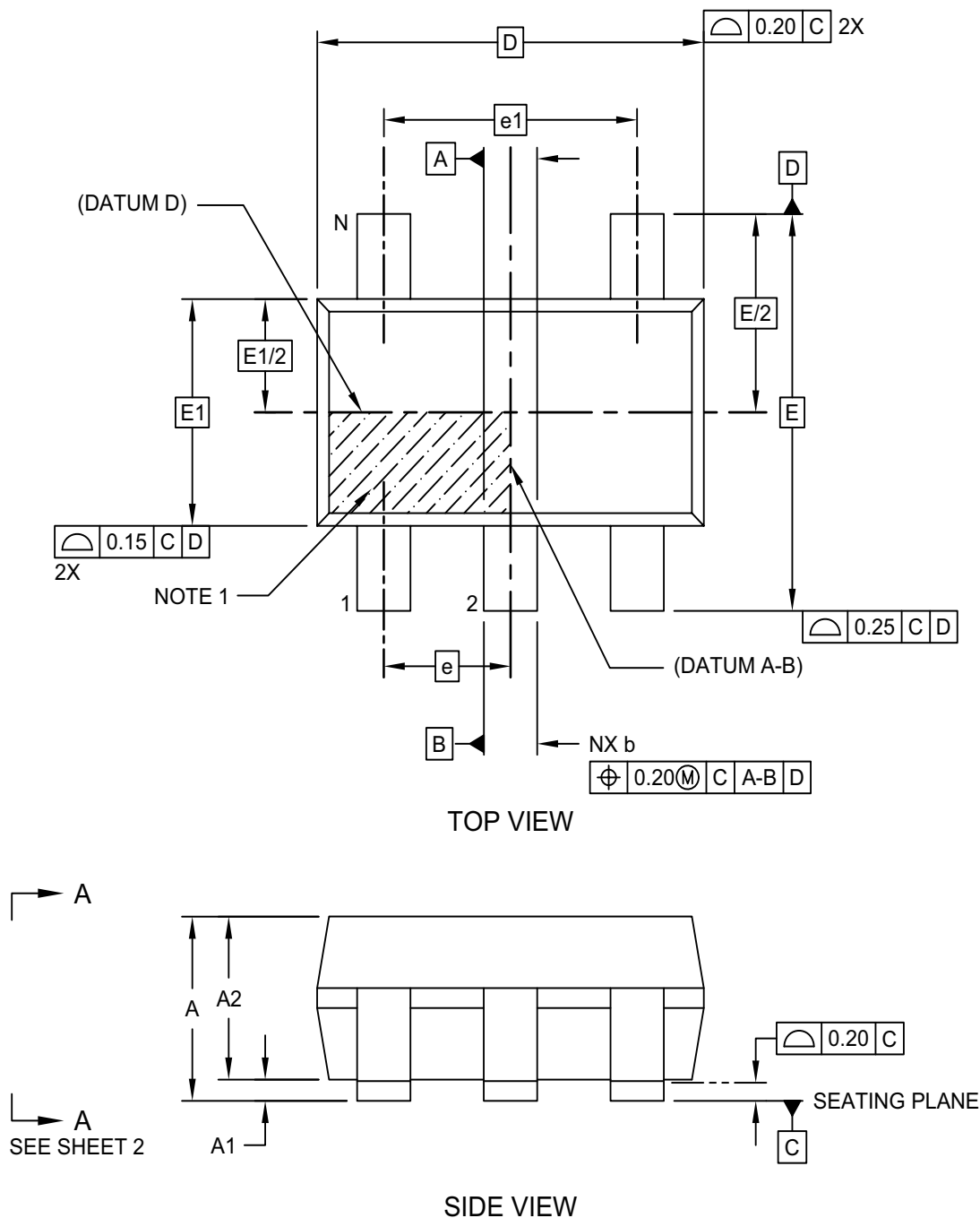
* This package is Pb-free. The Pb-free JEDEC designator () can be found on the outer packaging for this package.

Note: In the event the full Microchip part number cannot be marked on one line, it will be carried over to the next line, thus limiting the number of available characters for customer-specific information.

7.2 Package Drawings

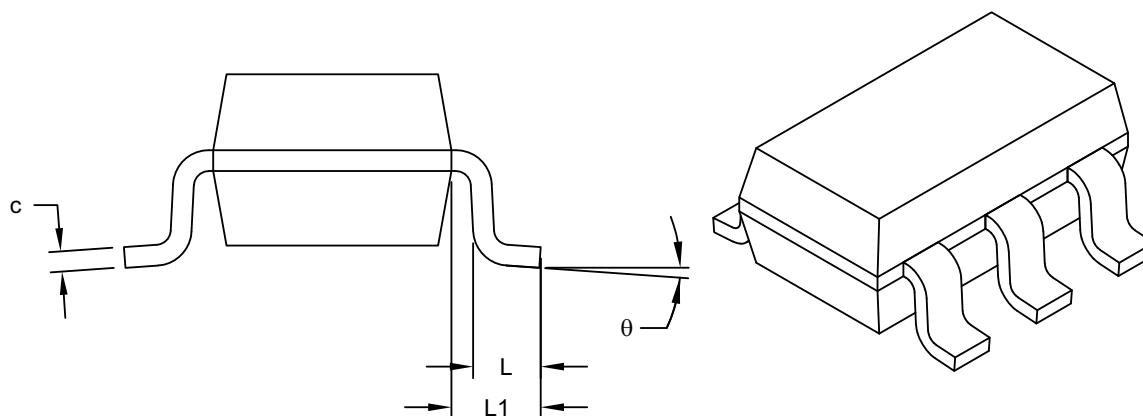
5-Lead Plastic Small Outline Transistor (OT) [SOT-23]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



5-Lead Plastic Small Outline Transistor (OT) [SOT-23]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



VIEW A-A
SHEET 1

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Pins	N	5		
Pitch	e	0.95 BSC		
Outside lead pitch	e1	1.90 BSC		
Overall Height	A	0.90	-	1.45
Molded Package Thickness	A2	0.89	-	1.30
Standoff	A1	-	-	0.15
Overall Width	E	2.80 BSC		
Molded Package Width	E1	1.60 BSC		
Overall Length	D	2.90 BSC		
Foot Length	L	0.30	-	0.60
Footprint	L1	0.60 REF		
Foot Angle	θ	0°	-	10°
Lead Thickness	c	0.08	-	0.26
Lead Width	b	0.20	-	0.51

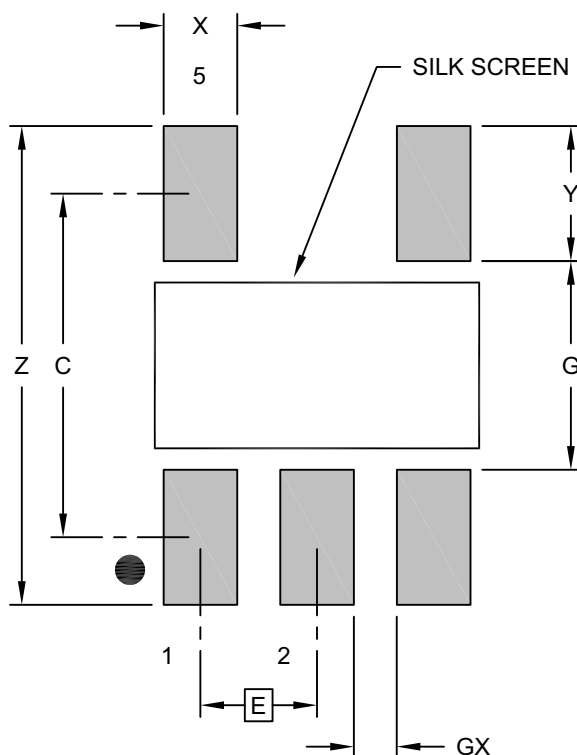
Notes:

- Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed 0.25mm per side.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-091-OT Rev H Sheet 2 of 2

5-Lead Plastic Small Outline Transistor (OT) [SOT-23]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.95 BSC		
Contact Pad Spacing	C		2.80	
Contact Pad Width (X5)	X			0.60
Contact Pad Length (X5)	Y			1.10
Distance Between Pads	G	1.70		
Distance Between Pads	GX	0.35		
Overall Width	Z			3.90

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing No. C04-2091-OT Rev H

APPENDIX A: REVISION HISTORY

Revision F (May 2025)

Following is the list of changes for this revision:

- Added AEC-Q100 Qualification to [Features](#) and [General Description](#).
- Added [Related Devices](#).
- Updated the temperature range from industrial (-40°C to +85°C) to extended (-40°C to +125°C) in [Section 1.0, "Electrical Specifications"](#).
- Updated [Table 1-1, "DC Electrical Specifications"](#) notes.
- Updated [Product Identification System](#).
- Added [Product Identification System \(Automotive\)](#).
- Changed terminology:
 - Master device → Host device
 - Slave device → Client device
- Updated [Section 7.1, "Package Marking Information"](#).
- Updated package drawings in [Section 7.2, "Package Drawings"](#).
- Several minor improvements to text formatting, parameter symbols and figures.
- Updated *Trademark*, *Legal Notice* and *Microchip Devices Code Protection Feature* information. For details, refer to the [Microchip Information](#) page.
- Removed **Worldwide Sales and Service** page.

Revision E (January 2017)

Following is the list of changes for this revision:

- Added a note to [Section 6.4.1, "Powering MCP3221"](#).
- Fixed the format in [Section 7.1, "Package Marking Information"](#) reflect the correct package marking as "1 2 3 4" instead of "1 10 10 10".
- Removed industrial temperature range information from [Table 1-2, "Temperature Specifications"](#).

Revision D (January 2013)

Following is the list of changes for this revision:

- Added a note to each package outline drawing.

Revision C (July 2006)

Following is the list of changes for this revision:

- Updated [Section 5.2, "Device Addressing"](#):
 - Changed 4-bit device code to 1001.
 - Changed the default value for the three address bits to 101.

Revision B (May 2003)

Following is the list of changes for this revision:

- Numerous changes throughout document.

Revision A (November 2002)

Initial release of this document.

NOTES:

PRODUCT IDENTIFICATION SYSTEM

To order or obtain information, e.g., on pricing or delivery, refer to the factory or the listed sales office.

<u>PART NO.</u>	<u>XX</u>	<u>X⁽²⁾</u>	<u>-X</u>	<u>/XX</u>	<u>XXX</u>																																																											
Device	Address	Tape and Reel	Temperature Range	Package	Class																																																											
Device: MCP3221 = 12-Bit ADC with I ² C Interface																																																																
<table> <tr> <th rowspan="2">Address Option:</th><th colspan="5">Address Bits</th></tr> <tr> <th><u>XX</u></th><th></th><th><u>A2</u></th><th><u>A1</u></th><th><u>A0</u></th></tr> <tr> <td>A0</td><td>=</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr> <td>A1</td><td>=</td><td>0</td><td>0</td><td>0</td><td>1</td></tr> <tr> <td>A2</td><td>=</td><td>0</td><td>1</td><td>1</td><td>0</td></tr> <tr> <td>A3</td><td>=</td><td>0</td><td>1</td><td>1</td><td>1</td></tr> <tr> <td>A4</td><td>=</td><td>1</td><td>0</td><td>0</td><td>0</td></tr> <tr> <td>A5⁽¹⁾</td><td>=</td><td>1</td><td>0</td><td>1</td><td>1</td></tr> <tr> <td>A6</td><td>=</td><td>1</td><td>1</td><td>1</td><td>0</td></tr> <tr> <td>A7</td><td>=</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> </table>						Address Option:	Address Bits					<u>XX</u>		<u>A2</u>	<u>A1</u>	<u>A0</u>	A0	=	0	0	0	0	A1	=	0	0	0	1	A2	=	0	1	1	0	A3	=	0	1	1	1	A4	=	1	0	0	0	A5 ⁽¹⁾	=	1	0	1	1	A6	=	1	1	1	0	A7	=	1	1	1	1
Address Option:	Address Bits																																																															
	<u>XX</u>		<u>A2</u>	<u>A1</u>	<u>A0</u>																																																											
A0	=	0	0	0	0																																																											
A1	=	0	0	0	1																																																											
A2	=	0	1	1	0																																																											
A3	=	0	1	1	1																																																											
A4	=	1	0	0	0																																																											
A5 ⁽¹⁾	=	1	0	1	1																																																											
A6	=	1	1	1	0																																																											
A7	=	1	1	1	1																																																											
Tape and Reel Option: T = Tape and Reel ⁽²⁾																																																																
Temperature Range: I = -40°C to +85°C (Industrial) E = -40°C to +125°C (Extended)																																																																
Package: OT = 5-Lead Plastic Small Outline Transistor, SOT-23																																																																
Class: Blank = Non-automotive VAO = Automotive																																																																
Note 1: Default Option. Contact Microchip factory for other address options. 2: Tape and Reel identifier only appears in the catalog part number description. This identifier is used for ordering purposes and is not printed on the device package. Check with your Microchip Sales Office for package availability with the Tape and Reel option.																																																																

Examples:

- a) MCP3221A0T-E/OT = 12-Bit ADC I²C Interface, Address A0, Tape and Reel, Extended Temperature, 5-Lead SOT-23
- b) MCP3221A1T-E/OT = 12-Bit ADC I²C Interface, Address A1, Tape and Reel, Extended Temperature, 5-Lead SOT-23
- c) MCP3221A2T-E/OT = 12-Bit ADC I²C Interface, Address A2, Tape and Reel, Extended Temperature, 5-Lead SOT-23
- d) MCP3221A3T-E/OT = 12-Bit ADC I²C Interface, Address A3, Tape and Reel, Extended Temperature, 5-Lead SOT-23
- e) MCP3221A4T-E/OT = 12-Bit ADC I²C Interface, Address A4, Tape and Reel, Extended Temperature, 5-Lead SOT-23
- f) MCP3221A5T-E/OT = 12-Bit ADC I²C Interface, Address A5, Tape and Reel, Extended Temperature, 5-Lead SOT-23
- g) MCP3221A5T-I/OT = 12-Bit ADC I²C Interface, Address A5, Tape and Reel, Industrial Temperature, 5-Lead SOT-23
- h) MCP3221A6T-E/OT = 12-Bit ADC I²C Interface, Address A6, Tape and Reel, Extended Temperature, 5-Lead SOT-23
- i) MCP3221A7T-E/OT = 12-Bit ADC I²C Interface, Address A7, Tape and Reel, Extended Temperature, 5-Lead SOT-23

MCP3221

PRODUCT IDENTIFICATION SYSTEM (AUTOMOTIVE)

To order or obtain information, e.g., on pricing or delivery, refer to the factory or the listed sales office.

<u>PART NO.</u>	<u>XX</u>	<u>X⁽²⁾</u>	<u>-X</u>	<u>/XX</u>	<u>XXX</u>																																																						
Device	Address	Tape and Reel	Temperature Range	Package	Class																																																						
Device: MCP3221 = 12-Bit ADC with I ² C Interface																																																											
Address Option: <table> <tr> <th></th><th><u>XX</u></th><th></th><th><u>A2</u></th><th><u>A1</u></th><th><u>A0</u></th></tr> <tr> <td>A0</td><td>=</td><td></td><td>0</td><td>0</td><td>0</td></tr> <tr> <td>A1</td><td>=</td><td></td><td>0</td><td>0</td><td>1</td></tr> <tr> <td>A2</td><td>=</td><td></td><td>0</td><td>1</td><td>0</td></tr> <tr> <td>A3</td><td>=</td><td></td><td>0</td><td>1</td><td>1</td></tr> <tr> <td>A4</td><td>=</td><td></td><td>1</td><td>0</td><td>0</td></tr> <tr> <td>A5⁽¹⁾</td><td>=</td><td></td><td>1</td><td>0</td><td>1</td></tr> <tr> <td>A6</td><td>=</td><td></td><td>1</td><td>1</td><td>0</td></tr> <tr> <td>A7</td><td>=</td><td></td><td>1</td><td>1</td><td>1</td></tr> </table>							<u>XX</u>		<u>A2</u>	<u>A1</u>	<u>A0</u>	A0	=		0	0	0	A1	=		0	0	1	A2	=		0	1	0	A3	=		0	1	1	A4	=		1	0	0	A5 ⁽¹⁾	=		1	0	1	A6	=		1	1	0	A7	=		1	1	1
	<u>XX</u>		<u>A2</u>	<u>A1</u>	<u>A0</u>																																																						
A0	=		0	0	0																																																						
A1	=		0	0	1																																																						
A2	=		0	1	0																																																						
A3	=		0	1	1																																																						
A4	=		1	0	0																																																						
A5 ⁽¹⁾	=		1	0	1																																																						
A6	=		1	1	0																																																						
A7	=		1	1	1																																																						
Tape and Reel Option: T = Tape and Reel ⁽²⁾																																																											
Temperature Range: I = -40°C to +85°C (Industrial) E = -40°C to +125°C (Extended)																																																											
Package: OT = 5-Lead Plastic Small Outline Transistor, SOT-23																																																											
Class: Blank = Non-automotive VAO = Automotive																																																											
Note 1: Default Option. Contact Microchip factory for other address options. Note 2: Tape and Reel identifier only appears in the catalog part number description. This identifier is used for ordering purposes and is not printed on the device package. Check with your Microchip Sales Office for package availability with the Tape and Reel option.																																																											
Examples: a) MCP3221A0T-E/OTVAO = 12-Bit ADC I²C Interface, Address A0, Tape and Reel, Extended Temperature, 5-Lead SOT-23, Automotive b) MCP3221A1T-E/OTVAO = 12-Bit ADC I²C Interface, Address A1, Tape and Reel, Extended Temperature, 5-Lead SOT-23, Automotive c) MCP3221A2T-E/OTVAO = 12-Bit ADC I²C Interface, Address A2, Tape and Reel, Extended Temperature, 5-Lead SOT-23, Automotive d) MCP3221A3T-E/OTVAO = 12-Bit ADC I²C Interface, Address A3, Tape and Reel, Extended Temperature, 5-Lead SOT-23, Automotive e) MCP3221A4T-E/OTVAO = 12-Bit ADC I²C Interface, Address A4, Tape and Reel, Extended Temperature, 5-Lead SOT-23, Automotive f) MCP3221A5T-E/OTVAO = 12-Bit ADC I²C Interface, Address A5, Tape and Reel, Extended Temperature, 5-Lead SOT-23, Automotive g) MCP3221A6T-E/OTVAO = 12-Bit ADC I²C Interface, Address A6, Tape and Reel, Extended Temperature, 5-Lead SOT-23, Automotive h) MCP3221A7T-E/OTVAO = 12-Bit ADC I²C Interface, Address A7, Tape and Reel, Extended Temperature, 5-Lead SOT-23, Automotive																																																											

Microchip Information

Trademarks

The “Microchip” name and logo, the “M” logo, and other names, logos, and brands are registered and unregistered trademarks of Microchip Technology Incorporated or its affiliates and/or subsidiaries in the United States and/or other countries (“Microchip Trademarks”). Information regarding Microchip Trademarks can be found at <https://www.microchip.com/en-us/about/legalinformation/microchip-trademarks>.

ISBN: 979-8-3371-1234-3

Legal Notice

This publication and the information herein may be used only with Microchip products, including to design, test, and integrate Microchip products with your application. Use of this information in any other manner violates these terms. Information regarding device applications is provided only for your convenience and may be superseded by updates. It is your responsibility to ensure that your application meets with your specifications. Contact your local Microchip sales office for additional support or, obtain additional support at www.microchip.com/en-us/support/design-help/client-support-services.

THIS INFORMATION IS PROVIDED BY MICROCHIP "AS IS". MICROCHIP MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND WHETHER EXPRESS OR IMPLIED, WRITTEN OR ORAL, STATUTORY OR OTHERWISE, RELATED TO THE INFORMATION INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF NON-INFRINGEMENT, MERCHANTABILITY, AND FITNESS FOR A PARTICULAR PURPOSE, OR WARRANTIES RELATED TO ITS CONDITION, QUALITY, OR PERFORMANCE.

IN NO EVENT WILL MICROCHIP BE LIABLE FOR ANY INDIRECT, SPECIAL, PUNITIVE, INCIDENTAL, OR CONSEQUENTIAL LOSS, DAMAGE, COST, OR EXPENSE OF ANY KIND WHATSOEVER RELATED TO THE INFORMATION OR ITS USE, HOWEVER CAUSED, EVEN IF MICROCHIP HAS BEEN ADVISED OF THE POSSIBILITY OR THE DAMAGES ARE FORESEEABLE. TO THE FULLEST EXTENT ALLOWED BY LAW, MICROCHIP'S TOTAL LIABILITY ON ALL CLAIMS IN ANY WAY RELATED TO THE INFORMATION OR ITS USE WILL NOT EXCEED THE AMOUNT OF FEES, IF ANY, THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THE INFORMATION.

Use of Microchip devices in life support and/or safety applications is entirely at the buyer's risk, and the buyer agrees to defend, indemnify and hold harmless Microchip from any and all damages, claims, suits, or expenses resulting from such use. No licenses are conveyed, implicitly or otherwise, under any Microchip intellectual property rights unless otherwise stated.

Microchip Devices Code Protection Feature

Note the following details of the code protection feature on Microchip products:

- Microchip products meet the specifications contained in their particular Microchip Data Sheet.
- Microchip believes that its family of products is secure when used in the intended manner, within operating specifications, and under normal conditions.
- Microchip values and aggressively protects its intellectual property rights. Attempts to breach the code protection features of Microchip product is strictly prohibited and may violate the Digital Millennium Copyright Act.
- Neither Microchip nor any other semiconductor manufacturer can guarantee the security of its code. Code protection does not mean that we are guaranteeing the product is “unbreakable”. Code protection is constantly evolving. Microchip is committed to continuously improving the code protection features of our products.